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(1) Extension Program/Project/Study Title:	
ABYAN KAGULANGAN Eco-Tourism Development: A Discovery Center for Conservation and Promotion of Philippine Native Trees	
Proponent Name:	Peter John D. Fantinalgo, MSAr
Academic Rank:	Asst. Prof. IV
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(2) Category (please check) ☐ Education and Skill Empowerment ☒ Healthcare and Environmental Stewardship ☐ Innovation in Sustainable Agri-fishery and Livelihood ☒ Video Presentation ☐ Poster	SDG Alignment ☐ No to poverty ☐ Zero hunger ☐ Good health and well-being ☐ Quality Education ☐ Gender Equality ☐ Clean water and sanitation ☐ Affordable and Clean Energy ☐ Decent work and economic growth ☒ Industry, Innovation and Infrastructure ☐ Reduced inequality ☒ Sustainable cities and communities ☐ Responsible consumption and production ☒ Climate action ☐ Life below water ☐ Life on land ☐ Peace, justice, and strong institutions ☐ Partnerships for the goals
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(4) Site of implementation	
Municipality/City:	Dumalag
Province:	Capiz
Region:	6

(5) Mode of Implementation	
☐ Single Authorship/Proponent ☐ Collaborative Extension (within campus) ☐ Collaborative Extension (intercampus) ☒ Collaborative Extension (external agency)	

(6) Sector/Commodity Secondary Sector (Design, Building and Infrastructure)	(7) Discipline/Field of Specialization Architecture (Green Design, Sustainable Architecture and Landscape Architecture)
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(8) Rationale (include the situation before intervention)

Philippine native trees serve an important part in conserving biodiversity and sustaining our ecosystem's delicate equilibrium. They provide critical nutrition and refuge for a wide variety of local animals, including birds, insects, and mammals. Planting native plants aids in the restoration and enhancement of the Philippines' natural environment. These trees developed to survive in the local environment, making them more resilient to shifting weather patterns and severe weather conditions in the country. With their prohibitive root systems, Philippine Native Trees promote soil quality, reduce erosion, and aid in water filtering. This improves nutrient cycling and offers a healthier environment for plants and microorganisms.

Unfortunately, many native trees are vanishing as a result of deforestation, replacement by invasive alien species, mono-crop plantations, and the effect of climate change. According to Wilson E. and Pang S., 2020 of the Department of Biological Sciences at the National University of Singapore, most dipterocarps' geographical ranges have already been decreased by more than 50% owing to deforestation. Model predictions integrating climate change scenarios until the year 2070 revealed that the combination of potential species range changes and current deforestation will significantly diminish the land area of future appropriate habitats for these tree species. This might bring numerous dipterocarp tree species closer together. The dual threats of deforestation and climate change may threaten an iconic tree family in the Philippines. The outcome of the study suggests incorporating climate change scenarios when planning for protected areas as environmentally suitable conditions may shift with climate change. Those species which are being guarded by protected areas now may no longer be there in the future, (Pang S.,2020).

Native trees have also been over-harvested and sold to other nations. As a result, many people are unfamiliar with our native trees but are more familiar with foreign species. The Philippines' fight against deforestation is ongoing, though progress has been achieved. The country established a variety of programs and regulations to encourage reforestation, sustainable forest management, and biodiversity protection. However, difficulties such as illicit logging and agricultural forest conversion continue.

The conservation status of the native species is being threatened. People need to understand that simply planting trees is not enough. The endemicity of the tree must be taken into consideration. Planting exotic trees bears no ecological significance. Worse even is that these exotic plants may compete with the natural floral for nutrients. At times, exotic flora may even completely overpower endemic ones which can be detrimental to our environment. With no proper information and distribution of communication with regards to the native trees, the community acquires back of knowledge in protecting and preserving it.

Conserving Philippine native trees is an intricate undertaking that necessitates the collaboration of government agencies, advocates for the environment, local communities, and individuals. It is critical to balance human growth with the need to safeguard and preserve the country's distinctive and valuable native tree species for the benefit of current and future generations.

(9)Project Description, Implementation date/duration, and Objectives

This project is a Physical Development Plan of the Proposed Discovery Center and its supporting facilities that promotes eco-tourism and conservation of Philippine Native Trees. This Development Plan was started on August 2023 and finished on June 2024. It was made through the collaboration of Capiz State University main campus BS Architecture Department with the collaboration of Capiz Provincial Environment and Natural Resources (CaPENRO).

Generally, this project aimed to design a Discovery Center with the necessary facilities that promote eco-tourism and conservation of Philippine Native Trees in Dumalag, Capiz.

Specifically, this study aimed to;

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1. to determine the spaces needed to conserve Philippine native trees, and eco-tourism, providing recreational stimulus to the public and stimulating the pleasure of learning;
2. to investigate the suitability of the site for the proposed Discovery Center in conserving and promoting Philippine native trees through the following criteria: site location, topography, site utilities, site vegetation, natural feature, and wildlife; and
3. to design a Discovery Center integrating biophilic design.

(10) Strategies employed (this may include activities to attain the objectives)

This study utilized qualitative research design to solve practical and specific problems. A case study that investigates a single subject, such as a person, group, location, event, organization, or phenomenon. It includes a in-depth interview for the participants, site observation and simulation to create solutions to architectural plans and digital 3D models. It is a procedure of imitative presentation in which a visualization of the structures is created to assess how it would operate in real-world circumstances.

The participants of this study was focused on tree conservation related professionals: Department of Environment and Natural Resources, Tourism Offices, Agriculture Professionals, Local Government Unit of Dumalag, Capiz, and other government agencies concerned who will contribute to ABYAN KAGULANGAN Eco-Tourism Development: A Discovery Center for Conservation and Promotion of Philippine Native Trees. This research proposal sought to determine the services and facilities required to be utilized by the users to provide a research facility to conserve Philippine Native Trees and promote tourism. The researchers have interviewed 12 participants who have in-depth knowledge and experience in the conservation measures of Philippine Native Trees and Tourism Management. These are foresters, agriculturists, tourism focal, the provincial government unit, and the local government unit of Dumalag, Capiz. The participants were chosen through purposive sampling.

Observation method was used to determine the suitability and characteristics to the site for this study. Observation includes a systematic and cautious observation of subjects, objects, or phenomena in their natural setting (Bhandari, 2021). Site visits to the place and the usage of recording equipment such as mobile phones and cameras to capture the conditions on the site will be utilized for gathering data. Understanding the context of the site will enable the researchers to identify challenges and potential solutions before the start of the research design phase. The locale of the study was in the province of Capiz, located at Sitio Kabanbanan, Brgy. Dolores, Dumalag, Capiz. Dolores is situated at approximately 11.2851, 122.6043, on the island of Panay. Elevation at these coordinates is estimated at 39.5 meters or 129.6 feet above mean sea level. The 15.00 hectares which is classified as Forestland, was selected for the study of the Discovery Center. The site has an irregular shape, is surrounded by both Forest Areas, and is situated along the well-known Mt. Paningraon of Dumalag, Capiz.

Simulation method used to create architectural solutions for the research, included floor plans, elevations, and digital models. It is a procedure of imitative presentation in which a visualization of the structures was created to assess how it would operate in real-world circumstances. The researchers have employ computer-aided technology to create a complete floor plan, elevations, sections, a digital 3D model, and a tour film (walkthrough) that will provide a sample of user experience on the intended building's model.

Research instruments used in this study included the interview guide and observation guide. Computer 3D and 2D software were utilized to make plans and simulations for the required translations. The 2D illustrations made using 'AutoCAD' that create scaled drawings to draw the exact dimensions and identify the adjacency of different spaces within the given measurements of the lot. Sketch-Up is a tool that is used to create 2D and 3D design models that allow us to visualize plans in a 3-dimensional view. Enscape has used to render and create a simulation to incorporate entourage, textures, materials, lighting, and sunlight into the building to analyze its effects and how it can be seen in the actual environment considering the site characteristics and its orientations. The final software as a post-

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production tool is 'Photoshop' to enhance the image quality and add additional elements to the rendered pictures.

(11) **Results** (extent of the accomplishment of the objectives) include unintended outcomes whenever available

Results of the Interview

The researchers conducted interviews with 12 respondents, including tourism officers, agriculturists, LGU employees, and foresters, to understand the establishment and management of a Philippine native tree discovery center, emphasizing biodiversity preservation, community engagement, research, and sustainable resource management.

The study found that establishing a Discovery Center for Philippine Native Trees offers benefits in conservation, education, rehabilitation, and promotion.

Participants highlighted the need for environmental facilities such as an arboretum, nursery, and wildlife rescue center, as well as educational facilities like a herbarium, research center, and information center at the Discovery Center.

They also emphasized the importance of conservation advocacy, innovative engagement, propagation purposes, and rehabilitation goals in ensuring long-term preservation of native tree species within the Eco-Tourism Development.

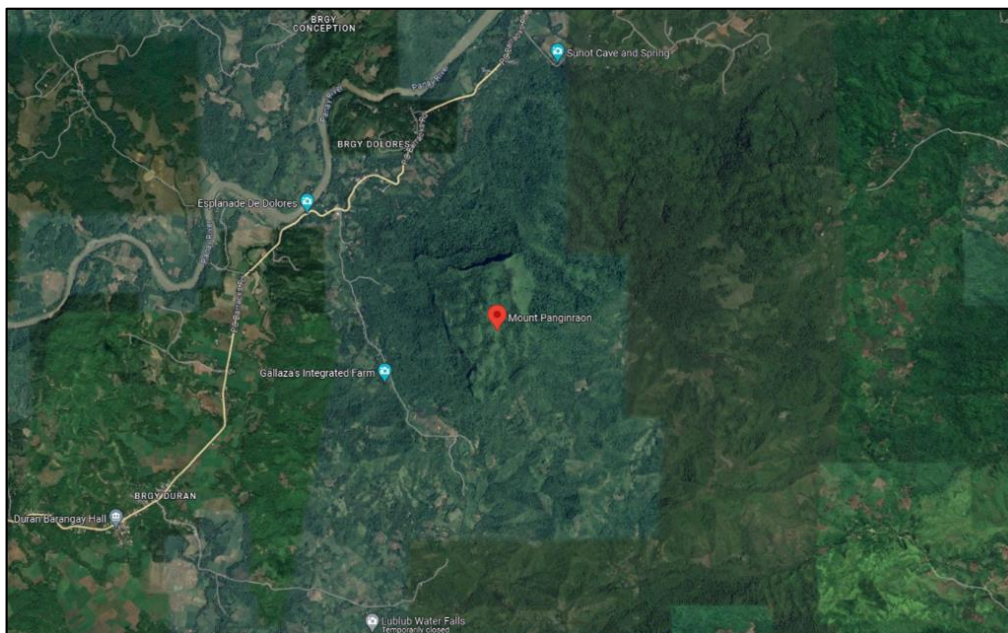
Activities such as tree planting, educational tours, and hiking, along with programs and exhibits focusing on education, conservation, recreation, exploration, and environmental engagement, were suggested to make learning about Philippine native trees at the Eco-Tourism Development more engaging.

Results of the Site Observation

The site for the Discovery Center is located in the forested area of Mount Panginra-on, offering diverse views of nearby towns and natural features.

The terrain features a flat foreground and a gradually ascending moderate slope, providing a varied and engaging landscape for visitors.

Access to the site is facilitated by a man made path that offers an exciting journey through the forest, enhancing the overall visitor experience.



Satellite Image of the Site Location

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Terrain Map of the Site Location



Topographic Map showing the Site Boundaries

The site benefits from the presence of water services from the Dumalag Water District and natural water reservoirs, ensuring a reliable water supply.

Electricity is supplied by the Capiz Electric Cooperative (CAPELCO), guaranteeing consistent and reliable power for residential and commercial purposes.

Communication facilities and network services are available from multiple providers, although internet connectivity may be intermittent or unreliable.

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View from Mt Panginraon



Site Small Water Stream



Site Water Utilities Pipe Line



Site Electrical Utilities Line

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The site's vegetation is characterized by a variety of tree species, including coconut, banana, and native species such as Teak, Hambabalud, and Takuyay Batino.

Coral-like rock formations and bodies of water, including small streams, add unique geological features to the site's landscape.

The area is several wildlife, with a diverse range of birds, reptiles, insects, and other animals contributing to its biodiversity



Site Vegetation (Running Bamboos)



Site Vegetation (Hambabalud Tree)



Site's Natural Feature (lime stone)



Site's Wildlife (Dupong Viper Snake)

SWOT Aanalysis

Strenght

The site's diverse flora and fauna, including endemic tree species and wildlife, enhance its ecological value and attractiveness to visitors. Natural rock formations, bodies of water, and dense forest cover contribute to the site's aesthetic appeal and provide diverse visual experiences. The site's natural features provide an opportunity for educational programs that raise awareness about conservation and biodiversity.

Weakness

The site's subtle variations in topography may pose challenges for construction and development, potentially increasing costs and timelines. Developing roads, parking, and facilities to

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support the Discovery Center may strain local resources and infrastructure. The site's rich biodiversity and natural features require careful planning to minimize negative impacts on the ecosystem during development.

Opportunity

The site's rich biodiversity and unique natural features makes it an attractive destination for eco-tourists seeking nature-based experiences. Opportunity to showcase environmental warriors and promote sustainable development. The Discovery Center could serve as a hub for research and conservation efforts focused on Philippine native trees and wildlife.

Threats

Habitat destruction: Development activities could lead to habitat destruction and fragmentation, impacting the site's biodiversity. **Climate change:** The site may be vulnerable to the effects of climate change, such as extreme weather events and sea-level rise, which could affect its ecological integrity.

Physical Development Plan of the Discovery Center for Conservation and Promotion of Philippine Native Trees



Aerial Perspective

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Site Development Plan

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For the Vedio Presentation (3D Simulation Walkthrough) click the link provided below:
https://drive.google.com/drive/folders/1CRURu8BQ54tpmxLP4R4mY3BAAXH_V7Vk



Parking Lot



Welcome Arch



Viewing Deck



Camping Area



Tree Signage



Arboretum



Trail

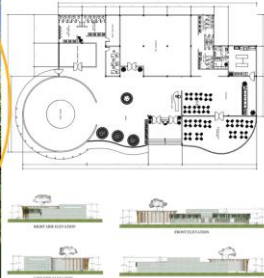


Resting Area

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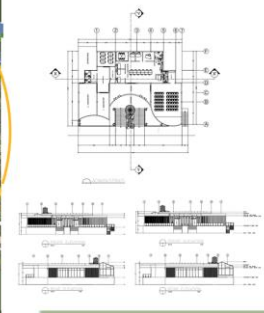
Main Building



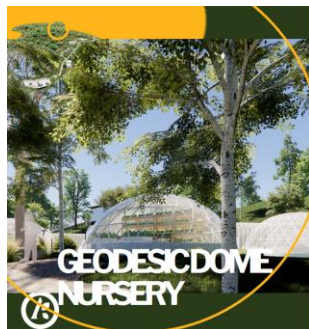
Main Building Interiors



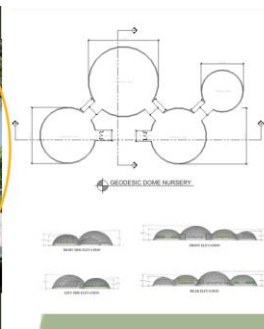
Administration Building



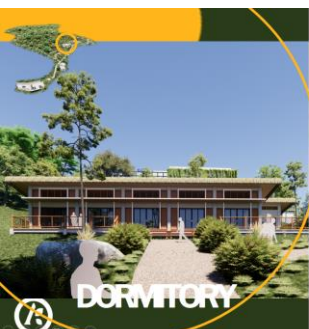
Administration Building Interiors



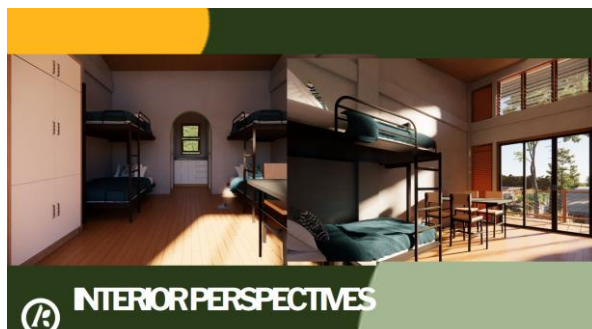
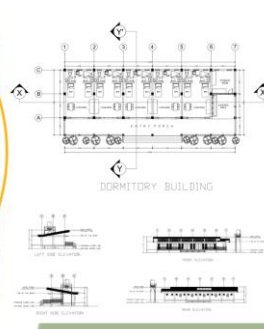
Geodesic Dome Nursery



Geodesic Dome Nursery Interiors



Dormitory



Dormitory Interiors

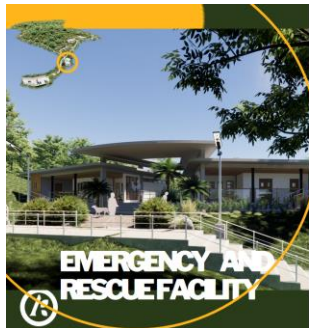
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Guest House



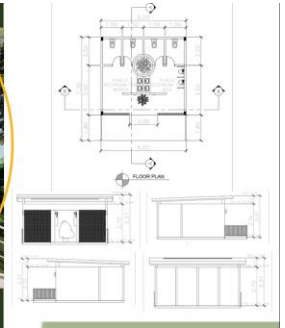
Guest House Interior



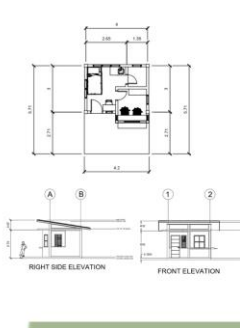
Emergency Rescue Facility



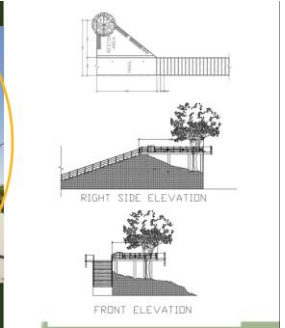
Public Restroom



Guard House



Resting Area



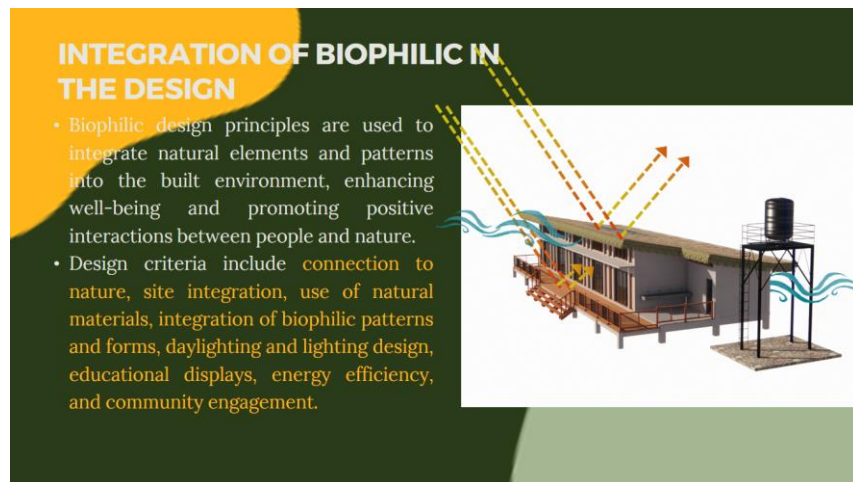
INTEGRATION OF BIOPHILIC IN THE DESIGN

- The design concept for the Discovery Center focuses on human and nature connection, minimizing environmental impact and providing a natural setting for users.
- It features elements like natural lighting, wide windows, and skylights, and incorporates trees into the building design, harmonizing with the environment.

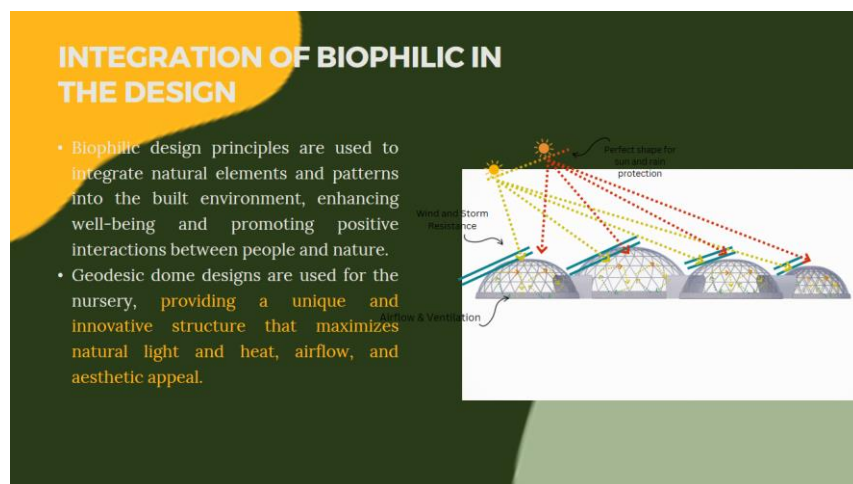


Biophilic Design Approaches (Natural Lighting and Large Windows)

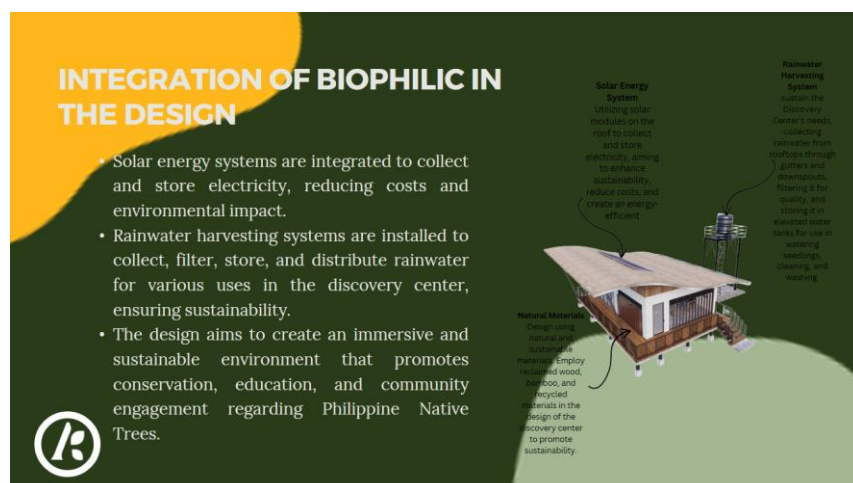
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Biophilic Design Approaches (Passive Cooling)



Biophilic Design Approaches (Protection Againsts the Elements)



Biophilic Design Approaches (Solar Energy System and Rain Water Harvesting system)

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(12) Lessons Learned/Insights Gained

The study emphasizes the importance of considering nature interaction in planning and designing spaces within the discovery center for the conservation and promotion of Philippine Native Trees. Community and environmental engagement, such as tree planting, educational tours, and hiking, should be integrated into the experience. The selected site at Sitio Kabanbanan, Brgy. Dolores, Dumalag, Capiz, was found to be ideal for the center's development. It is crucial to integrate biophilic design and sustainable architecture, including solar energy, rainwater harvesting, and natural lighting. The center should harmonize with the natural environment to provide a connection to nature.

(13) Recommendation/s

To the Local Government Unit (LGU), this study can be adopted by the province of Capiz to support the conservation and promotion of Philippine Native Trees and provide necessary facilities that can contribute to the development of the discovery center and the awareness of the public regarding Philippine Native Trees.

To the Environmental advocates, this study could offer a comprehensive approach to promoting the preservation and appreciation of Philippine native trees. The recommendations highlighted the significance of community engagement, environmental education, and sustainable practices. These insights can guide efforts to raise awareness about biodiversity conservation- especially Philippine Native Trees, and promotion, not just in Capiz but also in other regions that encounter similar conservation dilemmas.

To future researchers, this study could be used as a reference or guide by future researchers for those who are conducting in-line studies on the Discovery Center for Conservation and Promotion of Philippine Native Trees. This provides a solid foundation for future studies related to eco-tourism, conservation, and sustainable architecture.

The researchers suggest conducting a further study for an advanced development of the Discovery Center, implementing innovative technologies, and sustainable practices to enhance the overall sustainability and effectiveness of the Discovery Center.

(14) Acknowledgments (Include only important persons and institutions involved in funding and conducting the extension activities. Justify paragraphs with main headings in caps.)

This project stands as a testament to the collective effort and unwavering support of many individuals, without whom its completion would not have been possible.

First and foremost, to the Almighty God, whose divine guidance and blessings have illuminated and strengthened us through the path of researchers' journey. His wisdom has inspired and guided all who have played a role in bringing this manuscript to fruition.

Mr. Ronald Allan F. Cullo EnP, Acting CapEnRo, deserves special recognition for his unwavering support and guidance throughout the completion of this study. His contributions have been invaluable to the success of this research.

To all friends who provided unwavering assistance and support in various capacities during the research process, the researchers extend their deepest gratitude.

Special thanks to the respondents from various professions, as well as to CAPENRO, who actively participated in the research, contributing valuable insights and perspectives. The researchers also extend their gratitude to Mr. Japhet V. Bacanto and Kiua Marie Penuela for their invaluable support during the site visitation, providing documents, and processing the needs of the researchers. Additionally, the researchers are thankful to the province of Capiz for providing a vehicle every time they visit the site.

Lastly, to all others who have contributed in ways both seen and unseen, your support and assistance have been invaluable, and the researchers extend their deepest gratitude.

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